

Department of Computer Science and Engineering

PROJECT REPORT SUBMISSION

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Project Title: Smart Toll Collecting System with Object Detection

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Smart Toll Collecting System with Object Detection



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Abstract

Now a days there is a huge rush in the Toll plazas in order to pay the Toll Tax.Smart Toll collection system is used to minimize the time by collecting the toll amount electronically. To reduce the traffic jam and to save time we are introducing a IOT based Toll collection management system.

We have made the automation of Toll plaza using combination of microcontrollers. The aim of our project is to design a system which automatically identifies an approaching vehicles and notify to pay the tax and let the toll bar be opened after payment.If failed to pay the tax then it will receive a token with fixed fine according to the vehicle types.This system basically reduces the Traffic congestion at Toll plazas and helps in lower fuel consumption. This is very significant advantage of this system.

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Introduction

Electronic toll collection system allows the vehicle drives to pass the toll tax to booth without delaying.A toll is a payment made for the right to cross a gate or bridge or drive on a road. A toll can be managed publicly or privately.It generally associated with the fees charged for different modes of transportation (land, river, or sea) to have the right to travel on the respective roadway infrastructure: a car pays a toll to go onto a highway; a ship does so to cross a channel.Here we are going to create a smart Toll bar system with whom we can reduce traffic jam and create a user friendly Toll system which will be easy for every driver to pay tax and leave the Toll plaza easily. Normally in a Toll plaza the toll collecting man manually simulate the toll bar and also the same person collect toll tax from drivers which is very difficult for a man to do within a few seconds. That's why here we are introducing a automation system for toll plaza which will simulate a automation system with the help of IOT and it will be very easy to use.Besides IOT based system can keep the records of the vehicles.

Problem statement

We work here with a toll bar system where we combined IOT based system and object detection system . We drive car here manually but the toll plaza system is a automated system here. We work here for reduce traffic jam but the other project like this is mainly based on the paid system , but our project is based on toll positioning system of cars .

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Novelty of your project

Here we want to rebuild a great toll bar management system which will be very user friendly for the users.Our aim is to create a project which will lead to the next level from existing projects. Oue project is individual than other existing projects. Because we used microcontrollers, Arduinos, Drives, different types of modules, different types of sensors, node MCU ESP8266. Specially we try to make this project as IOT based project.We used dual Arduino, one for car & one for Toll booth. We used drive to run the car motors for dual wheels and also used Bluetooth module for to run the car with a Bluetooth connected device.Aftar that we used sonar sensor to detect the car in the Toll plaza and used a Bluetooth module for transaction.We also used node MCU ESP8266 for IOT so that we can make this project as IOT based.So by these features, we can say our project is individual than other existing projects.

Hardware Specifications

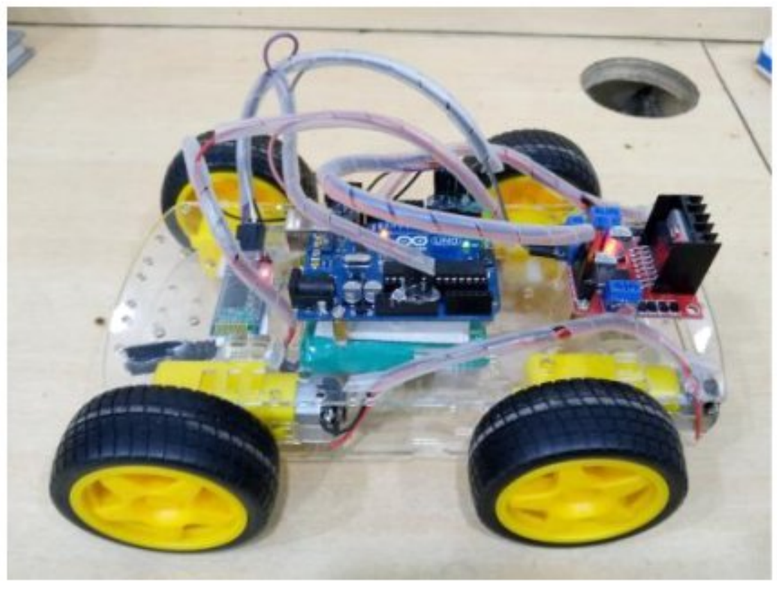
a. Arduino :



The **Arduino UNO** is the best board to get started with electronics and coding. If this is your first experience tinkering with the platform, the UNO is the most robust board you can start playing with. The UNO is the most used and documented board of the whole Arduino family.

b. Car Module ;

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Here we controll the car as our choice . Here we use arduino car module, bluettoth module and motor driver.

c. Bluettoth Module



Bluetooth module (Bluetooth module) refers to the basic circuit set of the chip with integrated Bluetooth function, used for short-range 2.4G wireless communication module. For the end user, the Bluetooth module is a semi-finished product

d. Motor Driver :



The **L298N** is a dual H-Bridge motor driver which allows speed and direction control of two DC motors at the same time. The module can drive DC motors that have voltages between 5 and 35V, with a peak current up to 2A. Let's take a closer look at the pinout of L298N module and explain how it works.

e. Ultra Sonic Sensor :

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As the name indicates, ultrasonic sensors measure distance by using ultrasonic waves. The sensor head emits an ultrasonic wave and receives the wave reflected back from the target**. Ultrasonic Sensors** measure the distance to the target by measuring the time between the emission and reception.Here we use it for object detection also.

f. Relay Module :



A power **relay module** is an electrical switch that is operated by an electromagnet. The electromagnet is activated by a separate low-power signal from a micro controller. When activated, the electromagnet pulls to either open or close an electrical circuit. Relay modules use low-level data signals to switch relays capable of handling loads up to 10 Amps

g. NodeMCU ESP8266 :

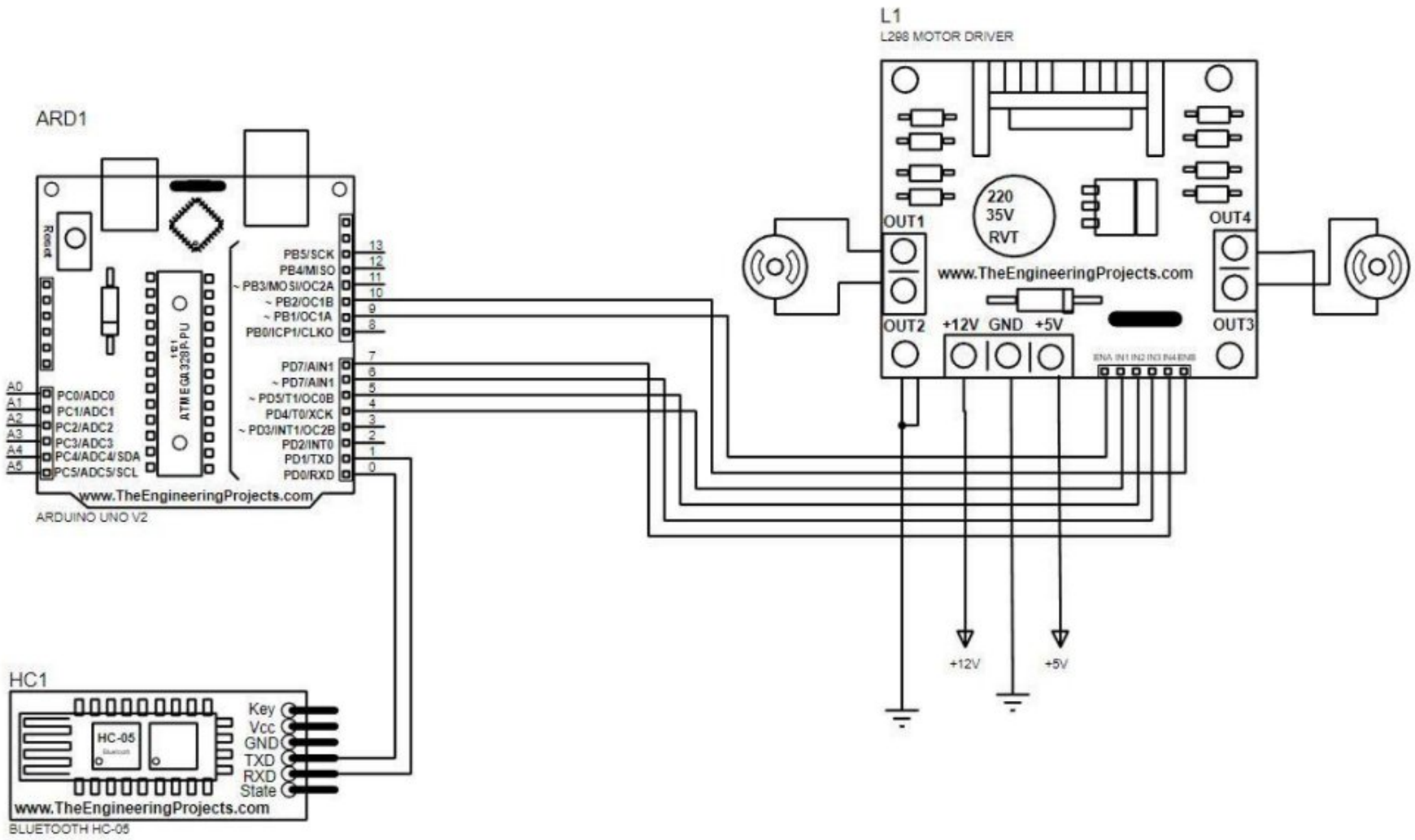


The **NodeMCU** (Node MicroController Unit) is an open-source software and hardware development environment built around an inexpensive System-on-a-Chip (SoC) called the **ESP8266**.We get Wifi support from it for a IOT output

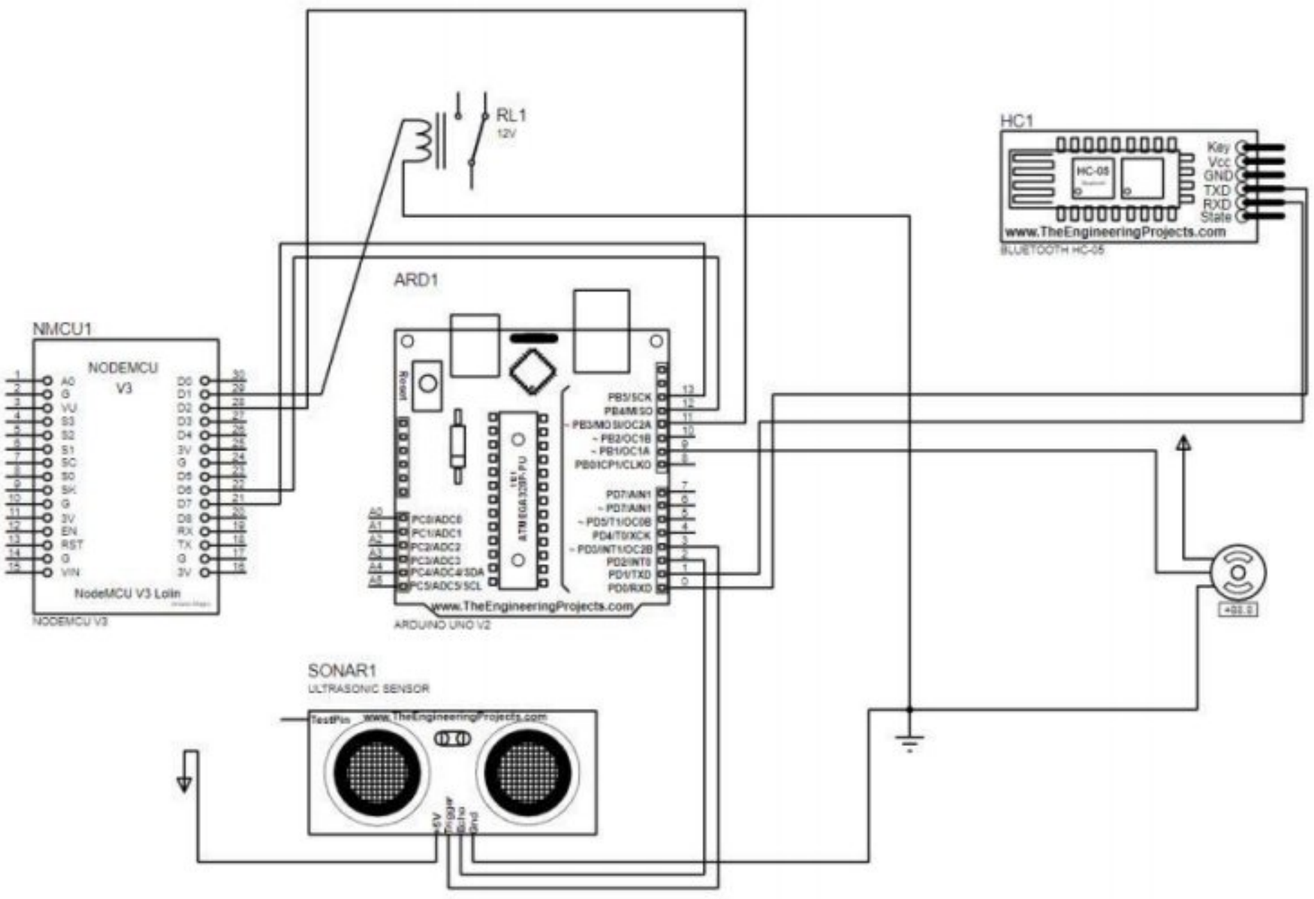
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Circuit Diagram

Car Diagram :



Automatic TollBar Diagram :



Project Prototype



Results

This project outcome will be helpful for both the government and it's people. Because government will able to get full revenue from Toll tax.Also there will no man need for to run the Toll booth because it will be IOT based system.

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Budget Statement

Arduino * 2 = 1100* 2	=	2200/=
Car Module *1 = 450	=	450/=
Bluettoth Mod *1 = 350	=	350/=
Motor Driver* 1 = 150	=	150/=
Ultra Sonic *1 = 100	=	100/=
Relay Module *1 = 100	=	100/=
Nodemcu * 1= 450	=	450/=
Jumper Cable*1 = 100	=	100/=
Servo Motor * 1 = 150	=	150/=
DC Battery * 4 = 100*4	=	600/=
Pencil Battery * 4 = 25*4	=	100/=
Charging Module 1* 100	=	100 /=
Extra	=	50/=
Total Cost	=	4900/ =

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Conclusion

We are trying to make a automatic smart Toll collection Booth which will detect vehichles,Receive a payment from vehicles,Give permission for to pass the Toll.People won't face traffic around the Toll Booth section.We can resolve corruption in this sector.No human error will occur as it is automatic Toll collection Booth.By this system we can reduce traffic jam around toll collection Booth, detect vehicle in toll tax place, detect paid or non paid vehicles,prevent Corruption, Reduce Operating Cost,develop a better management system in Toll Booth, maintain the time consumption during Toll collection.

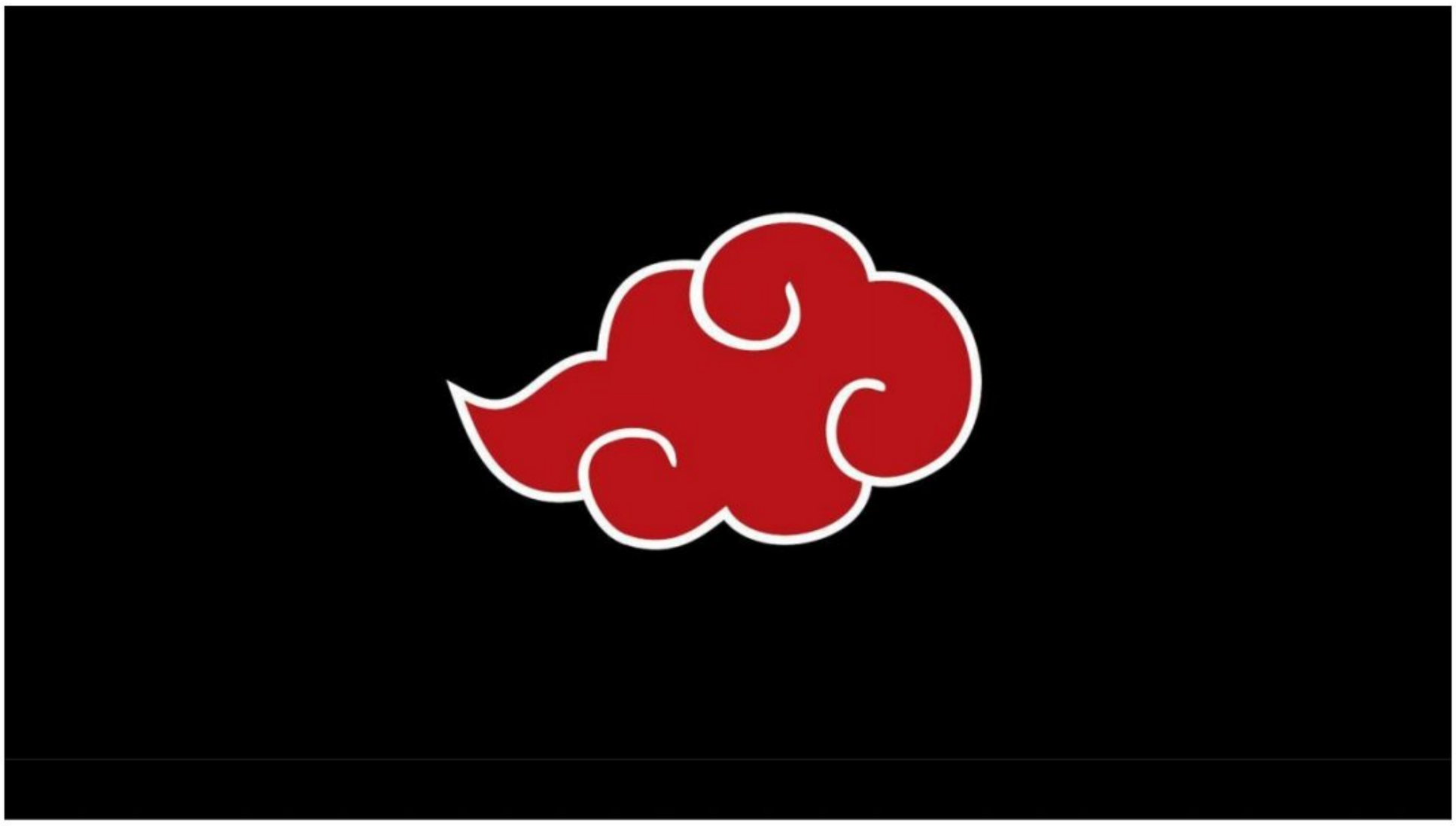
References

https://create.arduino.cc/projecthub

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TEAM

AKATSUKI



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